

# Brac University

Department of Electrical & Electronic Engineering

Semester Spring-25

Course Number: EEE103L

Course Title: Computer Programming Laboratory

Section: 06

## Lab Report

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Experiment no. 03 (Part 1)

Name of the experiment: Familiarization with Decision Control Structures & Loop Control Structures.

*Prepared by:*

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# Lab report of experiment 3

a. Any year is entered through the keyboard, write a program to determine whether the year is leap or not.

Code:

```
#include <stdio.h>

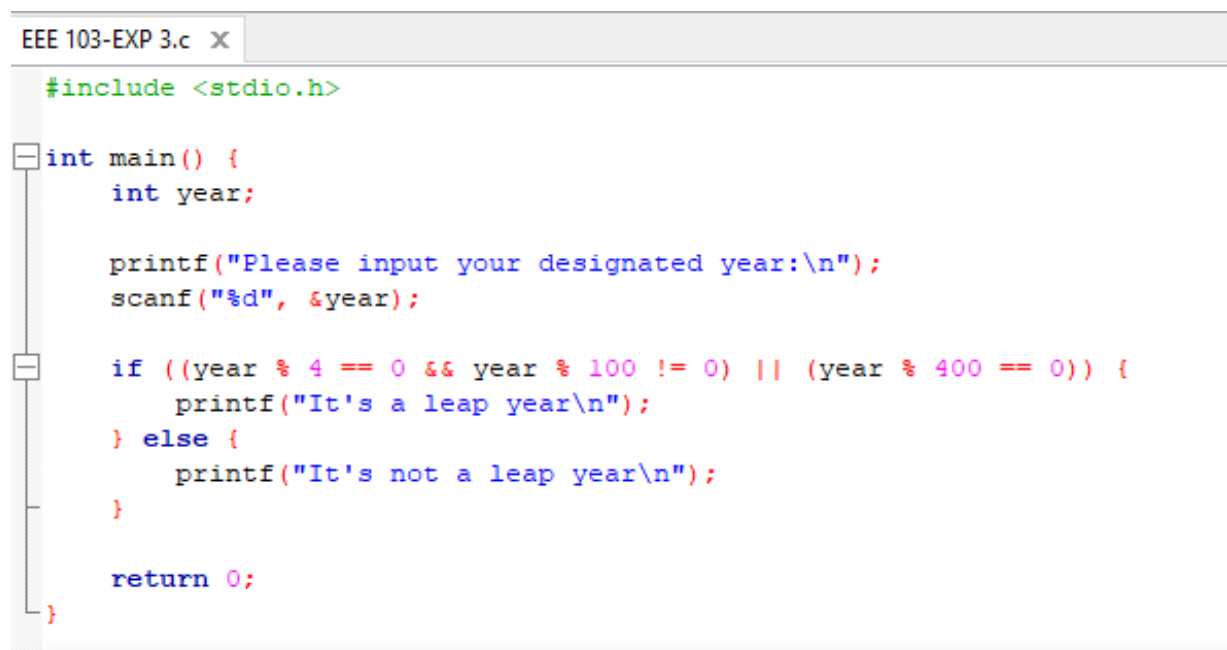
int main() {
    int year;

    printf("Please input your designated year:\n");
    scanf("%d", &year);

    if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
        printf("It's a leap year\n");
    } else {
        printf("It's not a leap year\n");
    }

    return 0;
}
```

Screenshot of code:

A screenshot of a code editor window titled 'EEE 103-EXP 3.c'. The code is written in C and is color-coded. It includes the standard input/output header, defines the main function, prompts the user for a year, reads the input, and uses a conditional statement to check if the year is a leap year based on the rules: divisible by 4 but not 100, or divisible by 400. The code is as follows:

```
#include <stdio.h>

int main() {
    int year;

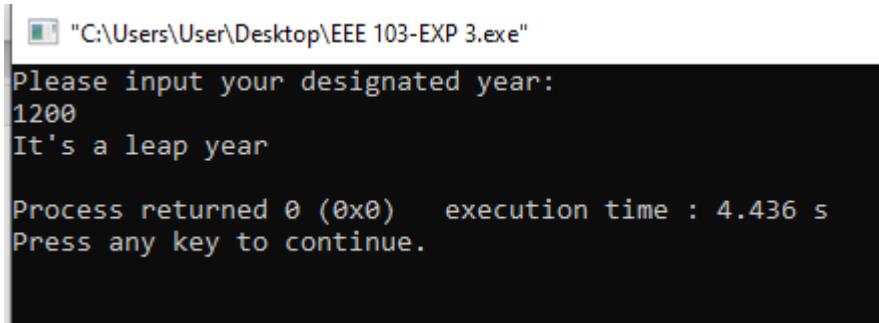
    printf("Please input your designated year:\n");
    scanf("%d", &year);

    if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
        printf("It's a leap year\n");
    } else {
        printf("It's not a leap year\n");
    }

    return 0;
}
```

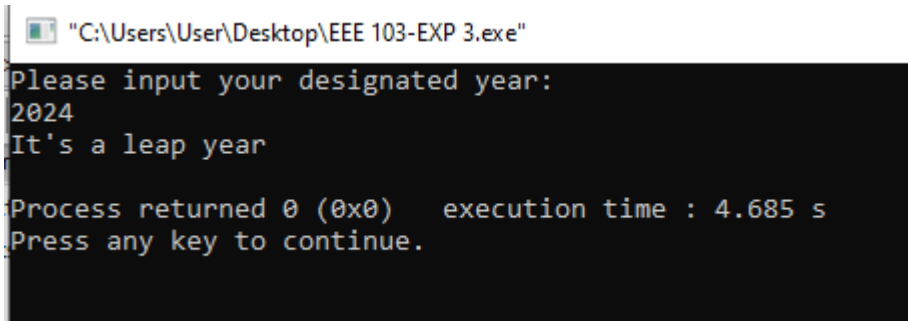
Output:

Leap year (1200, 2024)



```
"C:\Users\User\Desktop\EEE 103-EXP 3.exe"
Please input your designated year:
1200
It's a leap year

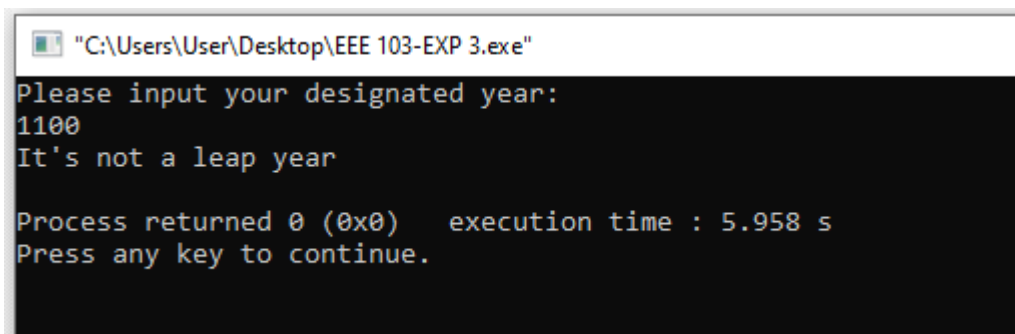
Process returned 0 (0x0)   execution time : 4.436 s
Press any key to continue.
```



```
"C:\Users\User\Desktop\EEE 103-EXP 3.exe"
Please input your designated year:
2024
It's a leap year

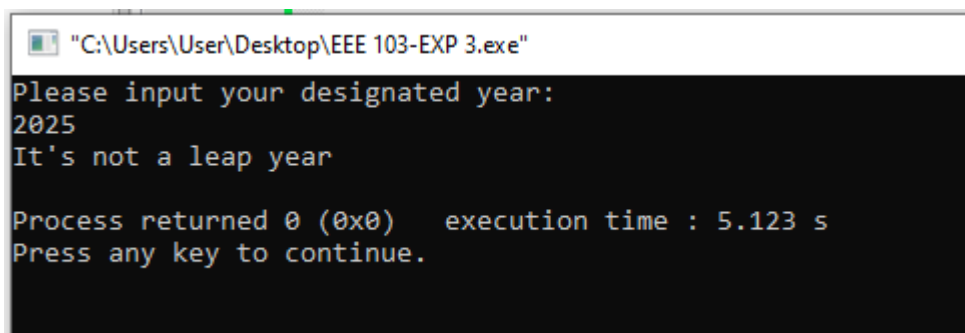
Process returned 0 (0x0)   execution time : 4.685 s
Press any key to continue.
```

Not a leap year (1100, 2025)



```
"C:\Users\User\Desktop\EEE 103-EXP 3.exe"
Please input your designated year:
1100
It's not a leap year

Process returned 0 (0x0)   execution time : 5.958 s
Press any key to continue.
```



```
"C:\Users\User\Desktop\EEE 103-EXP 3.exe"
Please input your designated year:
2025
It's not a leap year

Process returned 0 (0x0)   execution time : 5.123 s
Press any key to continue.
```

**b.** A certain grade of steel is graded according to the following conditions:

- (i) Hardness must be greater than 50
- (ii) Carbon content must be less than 0.7
- (iii) Tensile strength must be greater than 5600

The grades are as follows:

Grade is 10 if all three conditions are met

Grade is 9 if conditions (i) and (ii) are met

Grade is 8 if conditions (ii) and (iii) are met

Grade is 7 if conditions (i) and (iii) are met

Grade is 6 if only one condition is met

Grade is 5 if none of the conditions are met

Write a program, which will require the user to give values of hardness, carbon content and

tensile strength of the steel under consideration and output the grade of the steel.

Code:

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    float hardness, carbon, tensile;
```

```
    int cod1, cod2, cod3;
```

```
    printf("Enter the required formation amount of steel below\n");
```

```
    printf("Enter the hardness: ");
```

```
    scanf("%f", &hardness);
```

```
    printf("Enter the carbon content: ");
```

```
    scanf("%f", &carbon);
```

```
    printf("Enter the tensile strength: ");
```

```
    scanf("%f", &tensile);
```

```
    cod1= ( hardness>50 );
```

```
    cod2= ( carbon<0.7 );
```

```
    cod3= ( tensile>5600 );
```

```
    if(cod1 && cod2 && cod3)
```

```
        printf("Grade 10\n");
```

```
    else if (cod1 && cod2)
```

```
        printf("Grade 9\n");
```

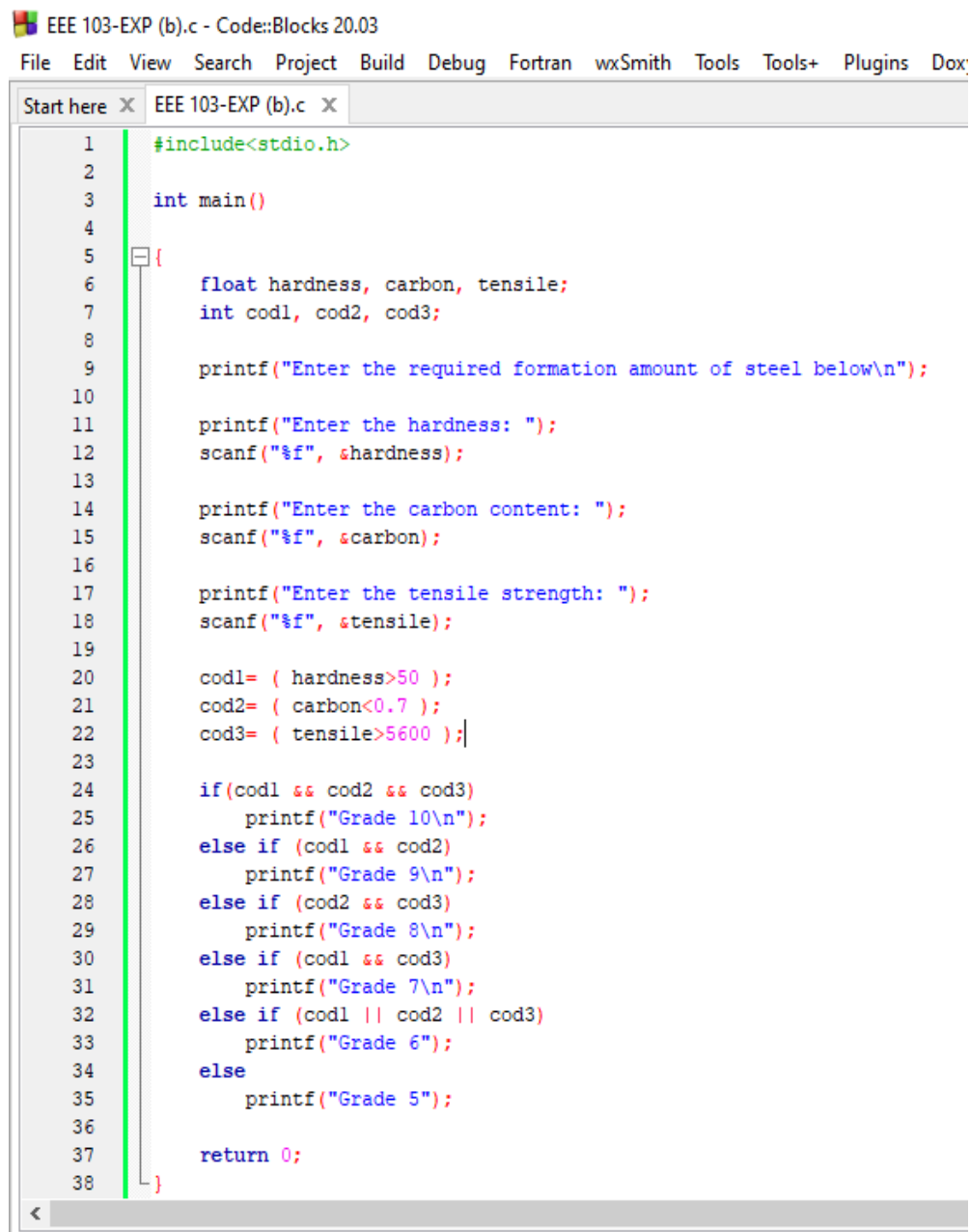
```

else if (cod2 && cod3)
    printf("Grade 8\n");
else if (cod1 && cod3)
    printf("Grade 7\n");
else if (cod1 || cod2 || cod3)
    printf("Grade 6");
else
    printf("Grade 5");

return 0;

```

Screenshot of code:



The screenshot shows a code editor window titled "EEE 103-EXP (b).c - Code::Blocks 20.03". The code is as follows:

```

1  #include<stdio.h>
2
3  int main()
4
5  {
6      float hardness, carbon, tensile;
7      int cod1, cod2, cod3;
8
9      printf("Enter the required formation amount of steel below\n");
10
11     printf("Enter the hardness: ");
12     scanf("%f", &hardness);
13
14     printf("Enter the carbon content: ");
15     scanf("%f", &carbon);
16
17     printf("Enter the tensile strength: ");
18     scanf("%f", &tensile);
19
20     cod1= ( hardness>50 );
21     cod2= ( carbon<0.7 );
22     cod3= ( tensile>5600 );
23
24     if(cod1 && cod2 && cod3)
25         printf("Grade 10\n");
26     else if (cod1 && cod2)
27         printf("Grade 9\n");
28     else if (cod2 && cod3)
29         printf("Grade 8\n");
30     else if (cod1 && cod3)
31         printf("Grade 7\n");
32     else if (cod1 || cod2 || cod3)
33         printf("Grade 6");
34     else
35         printf("Grade 5");
36
37     return 0;
38 }

```

Output:

Grade 10:

```
"C:\Users\User\Desktop\EEE 103\EEE 103-EXP (b).exe"
Enter the required formation amount of steel below
Enter the hardness: 55
Enter the carbon content: 0.5
Enter the tensile strength: 5700
Grade 10

Process returned 0 (0x0)   execution time : 28.985 s
Press any key to continue.
```

Grade 9:

```
Select "C:\Users\User\Desktop\EEE 103\EEE 103-EXP (b).exe"
Enter the required formation amount of steel below
Enter the hardness: 55
Enter the carbon content: 0.5
Enter the tensile strength: 5500
Grade 9
█
Process returned 0 (0x0)   execution time : 11.797 s
Press any key to continue.
```

Grade 8:

```
"C:\Users\User\Desktop\EEE 103\EEE 103-EXP (b).exe"
Enter the required formation amount of steel below
Enter the hardness: 45
Enter the carbon content: 0.5
Enter the tensile strength: 5700
Grade 8

Process returned 0 (0x0)   execution time : 8.953 s
Press any key to continue.
```

Grade 7:

```
"C:\Users\User\Desktop\EEE 103\EEE 103-EXP (b).exe"  
Enter the required formation amount of steel below  
Enter the hardness: 55  
Enter the carbon content: 1  
Enter the tensile strength: 5700  
Grade 7  
  
Process returned 0 (0x0)   execution time : 9.735 s  
Press any key to continue.
```

Grade 6:

```
"C:\Users\User\Desktop\EEE 103\EEE 103-EXP (b).exe"  
Enter the required formation amount of steel below  
Enter the hardness: 55  
Enter the carbon content: 3  
Enter the tensile strength: 5500  
Grade 6  
  
Process returned 0 (0x0)   execution time : 5.797 s  
Press any key to continue.
```

Grade 5:

```
"C:\Users\User\Desktop\EEE 103\EEE 103-EXP (b).exe"  
Enter the required formation amount of steel below  
Enter the hardness: 45  
Enter the carbon content: 4  
Enter the tensile strength: 4000  
Grade 5  
  
Process returned 0 (0x0)   execution time : 6.110 s  
Press any key to continue.
```